NOT MEASUREMENT SENSITIVE

MIL-DTL-32147
30 June 2004

DETAIL SPECIFICATION

SOCKS, FLYERS', ANTI-EXPOSURE, RUGGEDIZED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of flyers' ruggedized anti-exposure socks.

1.2 Classification. The socks are furnished in the following sizes as specified (see 6.2):

Size 1R – Women's boot size 3-4, right foot
Size 1L – Women's boot size 3-4, left foot
Size 2R – Women's boot size 5-6, right foot
Size 2L – Women's boot size 5-6, left foot
Size 3R – Women's boot size 7-8, right foot
Size 3L – Women's boot size 7-8, left foot
Size 4R – Women's boot size 9-10, right foot
Size 4L – Women's boot size 9-10, left foot
Size 5R – Men's boot size 6-7, right foot
Size 5L – Men's boot size 6-7, left foot
Size 6R – Men's boot size 8-9, right foot
Size 6L – Men's boot size 8-9, left foot
Size 7R – Men's boot size 10-12, right foot
Size 7L – Men's boot size 10-12, left foot
Size 8R – Men's boot size 13-15, right foot
Size 8L – Men's boot size 13-15, left foot

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division (Code 414100B120-3), Highway 547, Lakehurst, NJ 08733-5100 or emailed to Tom.Omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A FSC 8475

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

V-T-295 - Thread, Nylon

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-31011 - Cloth, Waterproof, and Moisture Vapor Permeable

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government drawings. The following Government drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of this document is that cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND

3769AS130 - Socks, Flyers', Ruggedized Anti-Exposure

(Copies of this drawing are available online at http://ct.dscp.dla.mil/contracting/, or from Defense Supply Center Philadelphia, Code CBTC Bldg. 6, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ-Z1.4	-	Procedures, Sampling, and Tables for Inspection by Attributes. (DoD adopted)

(Copies of this document are available from www.asq.org or the American Society for Quality, 600 Plankinton Avenue, Milwaukee, WI 53203.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) INTERNATIONAL

ASTM-D6193	-	Stitches and Seams, Standard Practice for. (DoD adopted)

(Copies of this document are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 (see 6.3).

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials and components. Unless otherwise specified in the contract or purchase order, the material and components shall not have been manufactured more than 18 months prior to the date of delivery of the sock(s).

3.3.1 Laminated cloth. The material for the socks shall be waterproof and moisture vapor permeable cloth conforming to MIL-DTL-31011, type IV, except the color shall be Green 3438.

3.3.2 Thread. The nylon thread for stitching the socks shall be in accordance with V-T-295, type I or II, size E. The color shall match the material specified in 3.3.1.

3.4 Design. The socks shall be tape sealed seamed with side and bottom panels.

3.5 Patterns. The Government furnished patterns (see 6.4) which show directional lines, and notches for assembly and seam allowances, shall not be altered in any way. The furnished

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Government patterns shall be used as a guide for cutting the working patterns. Minor modifications of the working patterns are permitted when using automated equipment or to meet a manufacturer's process, but the alterations shall not affect the serviceability, dimensions, or appearance of the socks.

3.5.1 List of pattern parts. The pattern parts and cutting quantities for the socks shall be as specified in table I.

TABLE I. Pattern parts

MATERIAL	PATTERN NOMENCLATURE	COMPUTER NOMENCLATURE	CUT PARTS
Laminate cloth	Sock Center Right	SOCK_CENTER_R	1
Laminate cloth	Sock Center Left	SOCK_CENTER_L	1
Laminate cloth	Sock Back and Sole Right	SOCK_BACK_R	1
Laminate cloth	Sock Back and Sole Left	SOCK_BACK_L	1

3.6 Construction. The socks shall be constructed in accordance with drawing 3769AS130 using sewn seams covered on the inside of the socks by heat sealed tape.

3.7 Marking. Cut parts shall be marked, ticketed, or bundled to ensure a uniform size throughout the socks. Any method of marking shall be used except metal fastening devices, sewn on tickets, or adhesive type tickets that leave traces of adhesive on the material after removal of the ticket.

3.8 Stitches, seams, and stitching. ASTM-D6193, Stitch Type 301, with 6 to 8 stitches per inch shall be used in the construction of the socks. Seams shall be as specified in the construction drawing (see 3.6).

3.9 Heat sealing. All sewn seams and stitching shall be covered on the inside of the socks and sealed with the seam tape. The entire width of the seam tape shall be heat sealed over the seams and stitching. Crossover areas and ends left unsealed after heat sealing shall be taped. To maintain a tape overlap, sealed sewn seams shall not contain more than two layers of seam tape per seam. The tape shall not lift from the seam at any point. End joints where the tape passes over taped seams on the socks shall be avoided. The overlap of splicings and tape ends shall be not less than ¾ inch.

3.10 Patching. A sock shall not have more than one patch. Patching of fabrication defects shall be permitted by a 2 inch length of seam tape on the inner side of the sock. Patches shall be applied using a heat sealer at a temperature of 350 ^{o}F for 25 seconds at 75 $\pm$ 2 foot-pounds

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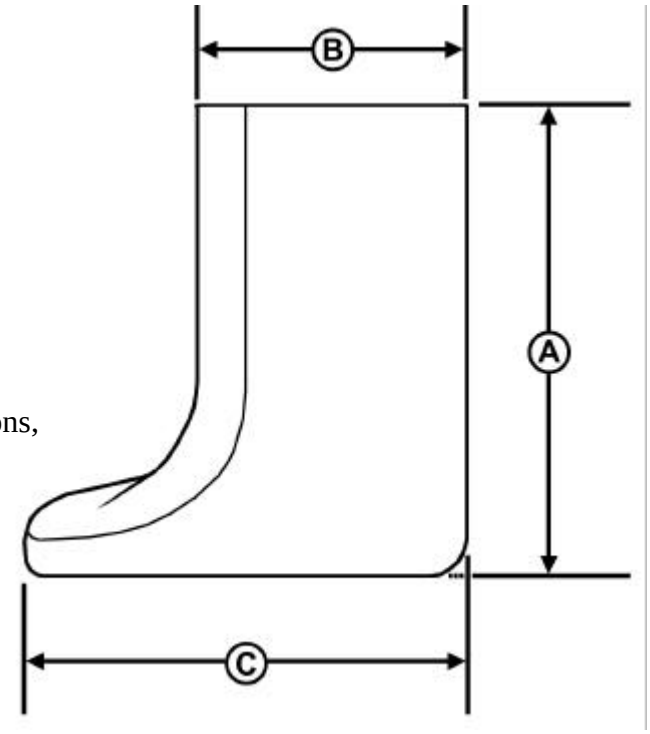
pressure. When patching is used, the patched area shall not leak as specified in 3.12.1. Major defects specified in table IV shall not be patched.

3.11 Finished dimensions. The finished socks dimensions shall be as specified in table II. A tolerance of ± 1/4 inch shall be used for all table II dimensions. The socks shall be measured as specified in 4.4.3 . Locations A, B, and C are shown on figure 1.

TABLE II. Finished dimensions (inches)

SIZE	HEIGHT A	WIDTH B	SOLE LENGTH C
1L, 1R	12 1/2	9	8 3/4
2L, 2R	12 1/2	9	9 1/4
3L, 3R	12 1/2	9	9 3/4
4L, 4R	12 1/2	9	10 1/4
5L, 5R	12 1/2	9	10 3/4
6L, 6R	12 1/2	9	11 1/4
7L, 7R	12 1/2	9	11 3/4
8L, 8R	12 1/2	9	12 1/4

.



Note:
1. Depicts table II dimensions, and is for informational purposes only.
2. Not to scale

FIGURE 1. Sock dimensions

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3.12 Performance requirements.

3.12.1 Sock leakage. All socks shall not leak and there shall not be any evidence of construction or material failure when tested as specified in 4.5.

3.13 Markings.

3.13.1 Identification. Each sock shall contain identification markings conforming to drawing 3769AS130.

3.14 Workmanship. The finished socks shall be uniform in quality and free from loose thread, foreign matter, and irregular defects that can adversely affect usage or durability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall consist of the examinations and tests specified in 4.4.1, 4.4.2, 4.4.3, 4.4.4, and 4.5.

4.2.1 First article samples. Unless otherwise specified in the contract or purchase order (see 6.2), the number of samples for first article inspections shall be one sock for each size ordered. The sample unit shall be one sock and the lot size shall be expressed in units of socks.

4.3 Conformance inspection. Conformance testing shall consist of the examinations specified in 4.4.1, 4.4.2, 4.4.3, 4.4.4, and 4.5.

4.3.1 Conformance inspection samples. Sampling for inspection shall be performed in accordance with ASQ-Z1.4. The sample unit shall be one sock and the lot size shall be expressed in units of socks.

4.4 Examinations and tests.

4.4.1 Material and component examinations and tests. Materials and components shall be examined and tested in accordance with the specified examinations and tests of table III.

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TABLE III. Material and component examinations and tests

MATERIALS	CHARACTERISTICS	REQUIREMENT PARAGRAPH	EXAMINATIONS AND TESTS
Laminated cloth	Material identification	3.3.1	All examinations and tests as specified in MIL-DTL-31011
Thread	Material identification	3.3.2	All examinations and tests as specified in V-T-295

4.4.2 Classification of end item defects. The end items shall be visually examined and measured for the defects specified in table IV.

TABLE IV. Classification of end item defects.

EXAMINE	DEFECT	CLASSIFICATION		
		CRITICAL	MAJOR	MINOR
General	Any hole, scissor or knife cut, tear, mend, burn, or weakening defect such as multiple floats, slubs, skips, needle chew or abraded area.		101	
	Any evidence of fabric delamination.		102	
	Any spot or stain (compound, oil, dirt, including marks) clearly visible.			201
	Color of any component not as specified.			202
	Any thread not trimmed to $^{1}/_{16}$ inch and not covered by heat sealed seam tape.		103	
	Any thread not trimmed to $^{1}/_{16}$ inch or thread scraps not removed.			203
	Any damaged tape edges greater than $^{1}/_{16}$ inch from outside edge.			204
	More than one 2-inch patch per sock.		104	
	Material more than 18 months old.	1		
Components and assembly	Any defective component or defect that will affect the form, fit, and function of the assembly.		105	
	Any component part or operation omitted, not as specified, or required operation improperly performed.		106	
	Any component part not cut in accordance with the patterns.		107	

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Table IV. Classification of end item defects – Continued.

EXAMINE	DEFECT	CLASSIFICATION		
		CRITICAL	MAJOR	MINOR
Heat sealing	Any seam tape not located as specified.		108	
	Any stitching not covered by seam tape.		109	
	Any seam tape not centered over stitched seam within 1/8 inch of the specified gauge.		110	
Seams and stitching	Any spliced tape not overlapped 3/4 inch.		111	
	Any seam or attachment of any component twisted, puckered, pleated or caught in any unrelated operation, or stitching that is not properly forced out or contained in a fold more than 1/8 inch.		112	
	End of stitching not securely backstitched for at least 1/2 inch when not caught in other seams or stitching.		113	
	Thread breaks, skips and run-offs not securely overstitched for at least 1/2 inch.		114	
	Any stitching irregular or unevenly gauged (greater than 50% of the seam length or 4 inches, whichever is less).		115	
	Not specified seam or stitch type.		116	
	Loose tension resulting in a loose seam or tight tension resulting in breaking of stitches when normal pull is applied.			205
	Stitches per inch - one or two stitches more or less than specified (to be scored only when condition exists on 25% of the seam or more).		117	
	Stitches per inch - more than two stitches more or less than specified (to be scored only when condition exists on 25% of the seam or more).		118	
Identification marking	Any open seam.		119	
	Identification missing, incorrect, or illegible.		120	
	Size marking missing, incorrect, or illegible.		121	
	Other than size information, markings incorrect, incomplete, or illegible.			206
	Misplaced by more than 1 inch.			207
End item measurements	Any dimension not within the specified tolerance.		122	

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4.4.3 End item dimensional examination. The end item shall be measured for conformance to the dimensions specified in 3.11 and table II as follows:

Dimension A - With the sock flat measure from the top of sock to the bottom of foot.

Dimension B - With the sock flat measure across the top of sock from folded edge to folded edge.

Dimension C - With the sock flat measure across the foot from toe to heel.

4.4.4 Visual examination. The end item shall be visually examined for compliance to 3.4, 3.6, 3.8, 3.9, 3.10, 3.13, and 3.14.

4.5 Sock leakage. Each assembled sock shall be tested for conformance to the requirements of 3.12.1. Any failure shall be cause for rejection of the sock. The completed sock shall be tested for waterproofness as follows:

a. Turn sock to be tested inside out. Care shall be taken to avoid excessive stretching which may damage the laminate material. Wet the outer taped surface with a soap solution of one liter of water to two drops of mild liquid soap. Be careful not to wet the inner untaped surface of the sock (face side of fabric).

b. Clamp the wet sock on a Gore sock tester (see 6.5). Inflate and maintain 0.5 psig air pressure.

c. To inspect for leakage, the sock shall be visually divided into 4 or 5 sections. Focusing on one small section of sock, thoroughly wet outer taped surface of that section, and allow seven to ten seconds to elapse while looking for formation of soap bubbles on surface of sock. Pay particular attention to edges of tape. Note the location of bubbles, respray, and as before allow seven to ten seconds to elapse while observing. Reappearance of soap bubbles at same location indicates a leak. Mark area of leakage with a china marker. Do not use ball point pen. If bubble formation is questionable, respray same area again waiting seven to ten seconds for bubbles to form. Do not repeat this procedure more than three times. If leakage is still questionable after three checks, dry sock completely before repeating these steps. Repeat this procedure on all other sock sections ensuring that the entire sock surface has been observed for leaks, including the sides and bottom.

d. Remove sock from tester. Dry sock thoroughly. Reseal leaking areas in accordance with 3.9 or patch in accordance with 3.10, as applicable. The sock shall be dry before using heat sealer. Do not heat seal when wet. Do not dry sock by applying heat with heat sealer. Damage which exceeds patching limits shall be cause for rejection of the sock.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or

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in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The socks are military unique due to the socks' use with the continuous wear coverall worn during overwater flights to protect the aircrewmember from hypothermia in the event of immersion in cold water. The socks are intended to be fitted on an individual basis by maintenance personnel for use with all sizes of continuous wear coveralls.

6.2 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification, including any amendments.
b. Size and quantity desired (see 1.2).
c. National stock number.
d. Applicable Government patterns and drawings, including revisions.
e. When first article is required (see 3.1).
f. First article samples (see 4.2.1).
g. Name and address of the first article inspection facility; and the name and address of the Government activity responsible for conducting the first article inspection program.
h. Packaging requirements (see 5.1).

6.3 First article. When first article inspection (see 3.1) is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Patterns. For access to patterns, contact the procuring activity issuing the invitation for bid.

6.5 Sources of supply. The Gore sock tester described in 4.5, is available from W. L. Gore & Associates, PO Box 729, Elkton, MD 21921.

6.6 Shelf-life. This specification covers items where shelf-life is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order. The shelf-life codes are contained in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M Shelf-life Management Manual, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points (ICPs), and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: http://www.shelflife.hq.dla.mil/.

6.7 Subject terms (key word) listing.
Anti-exposure coveralls
PTFE laminate stretch material
Sealing machine, heat
Shelf life

Custodians:
Army-AV
Navy-AS
DLA-CT

Preparing activity:
Navy - AS
(Project 8475-0019)

Review activity:
Navy-NU

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